

7.11 PreQuiz: Logarithmic and Exponential Functions — Markscheme

1. [Maximum mark: 6]

Maya invests \$1800 at 5 % per annum, compounded annually.

(a) End-of-year-1 value = $1800 \times 1.05 = 1890$. **AG**

(b) $V_8 = 1800(1.05)^8$ **M1**

$V_8 = \$2659.42$ **A1**

(c) Need the smallest integer n with $1800(1.05)^n > 2700$. **M1**

$n > \frac{\ln(2700/1800)}{\ln(1.05)} = 8.31 \dots$ **A1**

Smallest full-year $n = 9$ years. **A1**

2. [Maximum mark: 6]

$$F(t) = 1200e^{0.18t}.$$

(a) $F(1) = 1200e^{0.18} = 1436.66 \dots$, so about 1437 followers. **A1**

(b) $F(4) = 1200e^{0.72}$ **M1**

$F(4) = 2465.32 \dots$, so 2465 followers. **A1**

(c) Set $1200e^{0.18t} = 3000$. **M1**

$e^{0.18t} = 2.5$, so $0.18t = \ln 2.5$. **M1**

$t = \frac{\ln 2.5}{0.18} = 5.09$ years. **A1**